

Lynne J. Xu

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EDUCATION

Ph.D., Environmental Sciences

Expected June 2027

The University of California, Riverside • Riverside, CA

Research focuses on analyzing the pattern and influence of Southern California Air Quality

B.S., Environmental Sciences

The University of California, Riverside • Riverside, CA
Minor in Statistics

March 2022

Concentration in Natural Sciences

Engaged in air quality modeling research

RESEARCH EXPERIENCES

- Created hypothetical scenarios of identical fire conditions with different dates and corresponding meteorological conditions to analyze the difference in movement and dispersion of wildfire plumes. Created further statistical models to predict wildfire plumes' dispersion and exposure to nearby residents.
- Integrated computer modeled transportation-related emissions into the atmospheric chemical transport model, and analyzed the impact of modeled emissions
- Analyzed observational air quality data and examined the relative impact of local and long-range transport emissions, supported conclusion with modeled results

PROFESSIONAL APPOINTMENTS at UCR

Graduate Student Researcher

University of California, Riverside • Riverside, CA

June 2026 to Present

Teaching Assistant

University of California, Riverside • Riverside, CA

March 2026 to June 2026

Graduate Student Researcher

University of California, Riverside • Riverside, CA

June 2025 to March 2026

Teaching Assistant

University of California, Riverside • Riverside, CA

March 2025 to June 2025

Graduate Student Researcher

University of California, Riverside • Riverside, CA

June 2024 to March 2025

Teaching Assistant

University of California, Riverside • Riverside, CA

March 2024 to June 2024

Graduate Student Researcher

University of California, Riverside • Riverside, CA

January 2023 to March 2024

Research Assistance

University of California, Riverside • Riverside, CA

March 2022 to September 2022

Undergraduate Researcher

University of California, Riverside • Riverside, CA

November 2021 to March 2022

SELECTED FELLOWSHIP AND GRANTS

National Science Foundation (NSF) NSF Research Traineeship (NRT) SPEAKS Fellow

2023 - 2024

Science to Policy Education: Activating Knowledge for Sustainable Transportation (SPEAKS)

Includes training in sustainable transportation, public speaking, science to policy certificate

Work with numerous experts in different fields who are involved in policy-making

Dean's Distinguished Fellowship

Fall 2023

CONFERENCES

American Meteorological Society 106th Annual Meeting

January 2026

American Meteorological Society 105th Annual Meeting	January 2025
American Geophysical Union Fall Meeting	December 2023
American Geophysical Union Fall Meeting	December 2022
International Association of Wildland Fire	April 2022
American Geophysical Union Fall Meeting	December 2020

SELECTED POSTER PRESENTATIONS

American Meteorological Society 105th Annual Meeting	January 2025
American Geophysical Union Fall Meeting	December 2023
Informal Gathering on Atmospheric Science and Photochemistry	October 2023
Early Career Scientist Symposium	June 2023
American Geophysical Union Fall Meeting	December 2022
International Association of Wildland Fire	April 2022

SELECTED ORAL PRESENTATIONS

American Meteorological Society 106th Annual Meeting	January 2026
Seasonal Variability and Meteorological Drivers Affecting California Wildland Fire Emissions Transport and Human Health Impacts	
Analyzing the Meteorology Driven Spatial-temporal Variability in Transportation Air Quality Impacts on neighboring communities	

SOFTWARE SKILLS

- Rstudio: Integrated environment for R, a programming language for statistical analysis
- Weather Research and Forecasting - Chemistry: An atmospheric chemical transport model, which is widely used in the atmospheric science community
- AERMOD: An atmospheric dispersion model, commonly used by EPA and air quality management districts
- WRF-Stilt : a Lagrangian particle dispersion model (LPDM) for atmospheric transport with WRF informing meteorological data

- Python: A high-level, general-purpose programming language